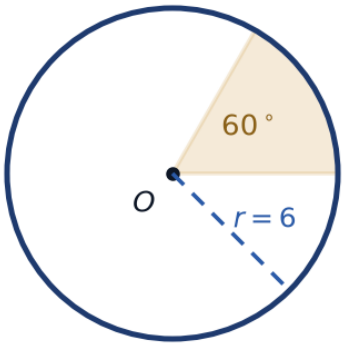
Functions: Modulus & Inverse

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

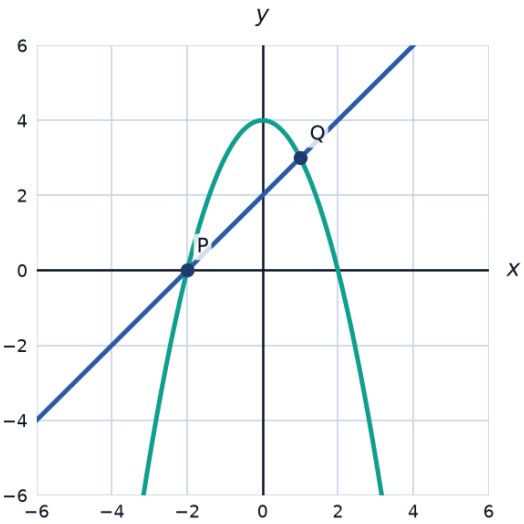
1 **[Foundational]** • **Target: Modulus Equations & Linear Graphs**

The complex number z is defined by $z = \frac{a+3i}{2+ai}$, where a is a real constant. Given that $\arg(z) = \frac{\pi}{4}$, find the value of a.



2 **[Foundational]** • **Target: Modulus Inequalities**

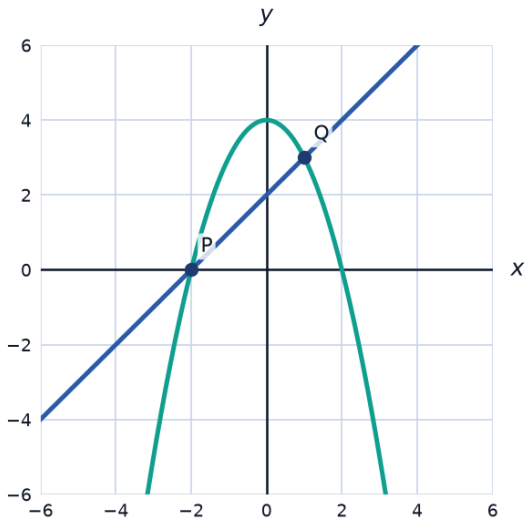
The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k$, where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|$. Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.



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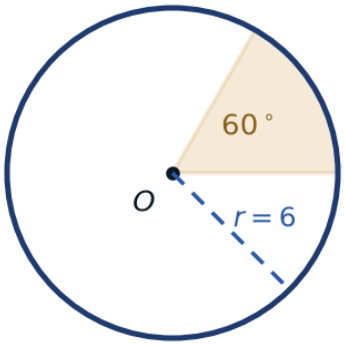
3 [Foundational] • Target: Composite and Inverse Functions in A2

The random variable X has a Poisson distribution with parameter λ, where $\lambda > 0$. Given that $P(X = 3) = P(X = 4)$, find the value of λ.



4 [Foundational] • Target: Transformations & Analytical Modeling

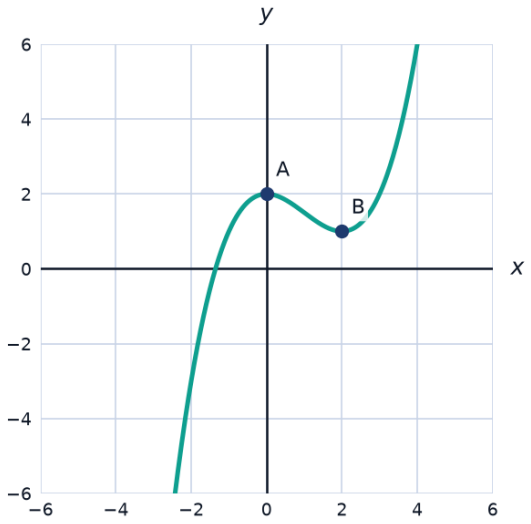
Given that $\cos\theta = -\frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$, find the exact value of $\sin 2\theta$.



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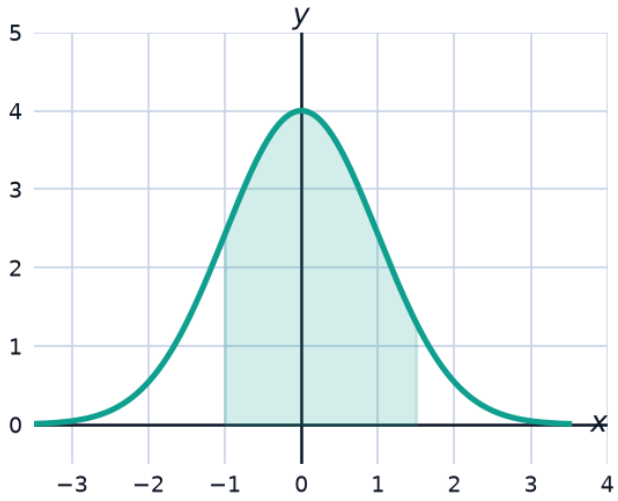
5 [Foundational] • Target: Modulus Equations & Linear Graphs

What are the values of x that satisfy the equation $|3x - 2| = 2|x + 1|$?



6 [Foundational] • Target: Modulus Inequalities

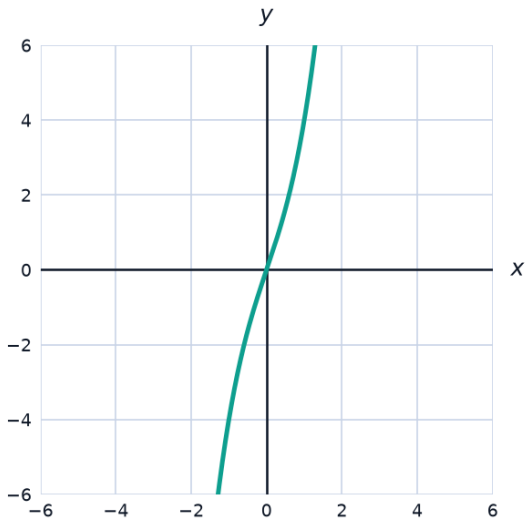
A machine produces steel rods whose lengths are normally distributed with standard deviation $\sigma = 2.4$ cm. A quality controller suspects that the mean length has increased from the target value of 50 cm. A random sample of 36 rods is measured, yielding a sample mean of $\bar{x} = 50.8$ cm. Find the value of the standardized test statistic z for testing $H_0 : \mu = 50$ against $H_1 : \mu > 50$.



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7 [Foundational] • Target: Composite and Inverse Functions in A2

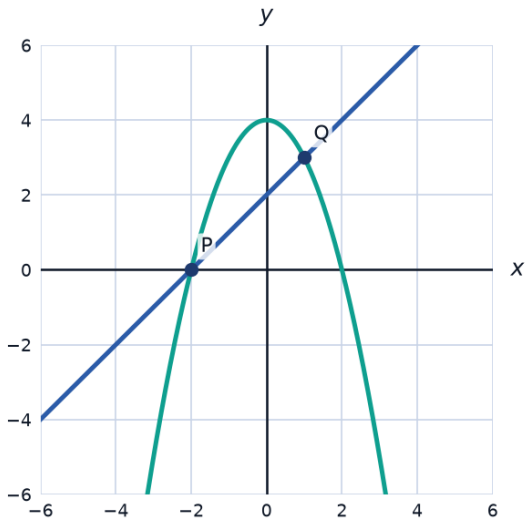
The function f is defined for all real numbers by $f(x) = x^3 + 3x - 5$. Between which pair of consecutive integers does the unique real root of the equation $f(x) = 0$ lie?



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Transformations & Analytical Modeling

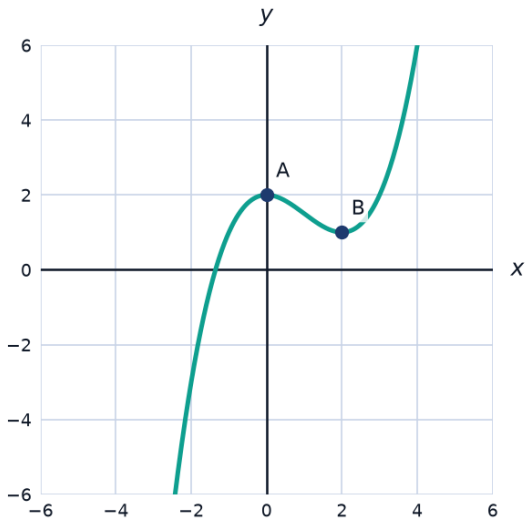
A continuous random variable X has probability density function given by $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, and $f(x) = 0$ otherwise. Find $P(1 \le X \le 2)$.



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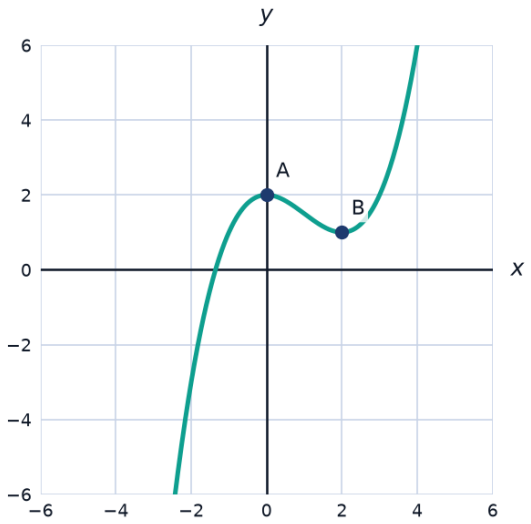
9 [Intermediate] • Target: Modulus Equations & Linear Graphs

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?



10 [Intermediate] • Target: Modulus Inequalities

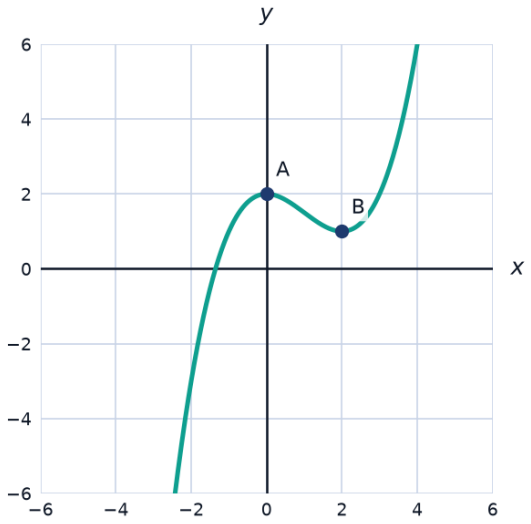
The first term of a geometric progression is 16 and the fourth term is 2. Find the sum to infinity, S_∞, of the progression.



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11 [Intermediate] • Target: Composite and Inverse Functions in A2

Evaluate $\int_0^3 2x(x^2+9)^4\,dx$.



12 [Intermediate] • Target: Transformations & Analytical Modeling

Find the set of all real solutions to the equation $\log_2(x-1) - \log_4(x+14) = 1$

13 [Intermediate] • Target: Modulus Equations & Linear Graphs

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

14 [Intermediate] • Target: Modulus Inequalities

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Composite and Inverse Functions in A2 • Cross-Lesson: 10.01 Numerical Methods

Calls arrive at a technical support desk according to a Poisson process with an average rate of λ calls per hour. It is known that the probability of receiving exactly 1 call in a given hour is 1.5 times the probability of receiving exactly 3 calls in that same hour. Find the probability that the desk receives exactly 4 calls during a randomly chosen 3-hour period.

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16 [Challenging] • Target: Transformations & Analytical Modeling • Cross-Lesson: 11.01 Differential Equations

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length 1 km and the second section has length 2 km. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X + Y = 3$, find the conditional probability $\mathrm{P}(X \geq 1 \mid X + Y = 3)$.

17 [Challenging] • Target: Modulus Equations & Linear Graphs • Cross-Lesson: 12.01 Poisson Distribution

A quality control process monitors the weight, W grams, of chocolate bars produced by a machine. A random sample of 50 chocolate bars was taken, and their weights were recorded. The sample mean weight was 102.5 grams, and the sample standard deviation was 3.5 grams. A) Calculate the standard error of the mean. B) Construct a 95% confidence interval for the mean weight of the chocolate bars. C) The machine is calibrated to produce chocolate bars with a mean weight of $\mu_0 = 100$ grams. Based on your confidence interval, what conclusion can be drawn about the calibration of the machine?

18 [Challenging] • Target: Modulus Inequalities • Cross-Lesson: 13.01 Continuous Random Variables

The curve C has equation $y = \frac{ax + 6}{x^2 + 3}$ where a is a constant. Given that C has a stationary point at $x = 1$, find the coordinates of the other stationary point of C.

19 [Challenging] • Target: Composite and Inverse Functions in A2 • Cross-Lesson: 14.01 Sampling & Estimation

The function f is defined for all $x \in \mathbb{R}$ by $f(x) = x^3 - 2x^2 - kx + 12$, where k is a real constant. The Newton-Raphson method is used to find an approximation to a root of the equation $f(x) = 0$. Given that an initial approximation $x_0 = 2$ produces a first approximation $x_1 = 4$, find the value of the constant k and the exact value of the second approximation x_2.

20 [Challenging] • Target: Transformations & Analytical Modeling • Cross-Lesson: 15.01 Hypothesis Testing

The complex number z is given by $z = \frac{4 + 3\mathrm{i}}{2 - \mathrm{i}}$. Express z in the Cartesian form $x + \mathrm{i}y$, where x and y are real numbers.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Modulus Equations & Linear Graphs

Solve the modulus equation $|3x - 5| = 2x + 1$ for all real solutions.

a. $x = 6$ and $x = 0.8$

b. $x = 6$ only

c. $x = 4$ and $x = -2$

d. $x = 0.8$ only

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22 [Foundational] • Target: Modulus Equations & Linear Graphs

Solve the modulus equation $|2x - 7| = |x + 2|$.

a. $x = 9$ and $x = 5/3$

b. $x = 9$ only

c. $x = 5$ and $x = -3$

d. $x = 3$ and $x = -1$

23 [Foundational] • Target: Modulus Equations & Linear Graphs

Find the coordinates of the vertex of the graph $y = |2x - 6| + 4$.

a. $(3, 4)$

b. $(-3, 4)$

c. $(3, -4)$

d. $(6, 4)$

24 [Foundational] • Target: Modulus Inequalities

Solve the inequality $|2x - 5| < 7$.

a. $-1 < x < 6$

b. $x < -1$ or $x > 6$

c. $-6 < x < 1$

d. $1 < x < 6$

25 [Foundational] • Target: Modulus Inequalities

Solve the inequality $|3x + 1| \geq |x - 4|$.

a. $x \leq -5/2$ or $x \geq 3/4$

b. $-5/2 \leq x \leq 3/4$

c. $x \leq -1$ or $x \geq 4$

d. $x \geq 3/4$

26 [Foundational] • Target: Modulus Inequalities

Find the set of values of x satisfying $|x - 2| > 2x - 1$.

a. $x < 1$

b. $x > 1$

c. $-1 < x < 1$

d. $x < 3$

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27 [Foundational] • Target: Composite and Inverse Functions in A2

The function $f(x) = 2 + \ln(x - 3)$ is defined for $x > 3$. Find the inverse function $f^{-1}(x)$ and state its domain.

a. $f^{-1}(x) = 3 + e^{x-2}$, domain $x \in \mathbb{R}$

b. $f^{-1}(x) = 3 - e^{x-2}$, domain $x > 0$

c. $f^{-1}(x) = 2 + e^{x-3}$, domain $x \in \mathbb{R}$

d. $f^{-1}(x) = \ln(x - 2) + 3$, domain $x > 2$

28 [Foundational] • Target: Composite and Inverse Functions in A2

Given $f(x) = e^{2x}$ for $x \in \mathbb{R}$ and $g(x) = \frac{1}{x}$ for $x > 0$, solve the equation $fg(x) = e^4$.

a. $x = 1/2$

b. $x = 2$

c. $x = 1/4$

d. $x = 4$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Composite and Inverse Functions in A2

A function $h(x) = x^2 - 4x + 1$ is defined on the domain $x \leq k$. Find the greatest value of k for which h is one-to-one.

a. $k = 2$

b. $k = 4$

c. $k = 0$

d. $k = 1$

30 [Intermediate] • Target: Transformations & Analytical Modeling

Describe the transformation mapping the graph of $y = \ln(x)$ onto the graph of $y = \ln(3x - 6)$.

a. Horizontal stretch scale factor $1/3$ followed by translation $[2, 0]$

b. Translation $[6, 0]$ followed by horizontal stretch scale factor 3

c. Horizontal stretch scale factor 3 followed by translation $[-2, 0]$

d. Vertical stretch scale factor 3 followed by translation $[2, 0]$

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31 [Intermediate] • Target: Transformations & Analytical Modeling

Find the number of real roots of the equation $|2x - 3| = x^2 - 2$.

a. 2 roots

b. 4 roots

c. 0 roots

d. 1 root

32 [Intermediate] • Target: Transformations & Analytical Modeling

The graph of $y = f(x)$ passes through $(2, 5)$. What are the coordinates of the corresponding point on $y = 3 - 2f(x + 1)$?

a. $(1, -7)$

b. $(3, -7)$

c. $(1, 13)$

d. $(3, 7)$

33 [Intermediate] • Target: Modulus Equations & Linear Graphs

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ or $5 < k < 11$

b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

34 [Intermediate] • Target: Modulus Inequalities

The function f is defined for $x \in \mathbb{R}, x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $x \neq 2$?

a. $\frac{3x+1}{x-2}$

b. $\frac{x-3}{2x+1}$

c. $\frac{3x-1}{x+2}$

d. $\frac{x+1}{2x-3}$

35 [Intermediate] • Target: Composite and Inverse Functions in A2

Given $f(x) = x + 2$ and $g(x) = 2x + 3$, find $fg(-4)$.

a. -5

b. -2

c. -3

d. -1

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36 [Intermediate] • Target: Transformations & Analytical Modeling

Given $f(x) = -3x - 11$, evaluate $f(|-6|)$.

a. -29

b. -26

c. 29

d. 7

37 [Intermediate] • Target: Modulus Equations & Linear Graphs

Given $f(x) = -3x - 1$ and $g(x) = -4x + 13$, find $fg(0)$.

a. -1

b. 13

c. 17

d. -40

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Modulus Inequalities • Cross-Lesson: 13.01 Continuous Random Variables

The function $f(x) = (x + 11)^2 - 4$ has domain $x \geq -11$. State the range of f.

a. $f(x) \leq -4$

b. $f(x) \geq -4$

c. $f(x) \geq -11$

d. $x \geq -11$

39 [Challenging] • Target: Composite and Inverse Functions in A2 • Cross-Lesson: 14.01 Sampling & Estimation

Given $f(x) = 2x + 3$ and $g(x) = 3x + 3$, find $fg(4)$.

a. 33

b. 36

c. 11

d. 15

40 [Challenging] • Target: Transformations & Analytical Modeling • Cross-Lesson: 15.01 Hypothesis Testing

The function $f(x) = (x + 8)^2 - 10$ has domain $x \geq -8$. State the range of f.

a. $f(x) \geq -8$

b. $f(x) \geq -10$

c. $f(x) \leq -10$

d. $x \geq -8$

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41 [Challenging] • Target: Modulus Equations & Linear Graphs • Cross-Lesson: 2.01 Exponentials & Logarithms

Given $f(x) = 3x + 8$, evaluate $f(|-5|)$.

a. 23

b. 26

c. -7

d. -23

42 [Challenging] • Target: Modulus Inequalities • Cross-Lesson: 3.01 Advanced Trigonometry

Given $f(x) = 5x + 8$, evaluate $f(|-9|)$.

a. 53

b. -53

c. -37

d. 56

43 [Challenging] • Target: Composite and Inverse Functions in A2 • Cross-Lesson: 4.01 Differentiation

Given $f(x) = -4x - 11$, evaluate $f(|6|)$.

a. 35

b. -35

c. -31

d. -32

44 [Challenging] • Target: Transformations & Analytical Modeling • Cross-Lesson: 5.01 Integration

Given $f(x) = -2x - 9$, evaluate $f(|4|)$.

a. -17

b. 17

c. -14

d. -13

45 [Challenging] • Target: Modulus Equations & Linear Graphs • Cross-Lesson: 6.01 Sequences & Series

Given $f(x) = 3x + 1$ and $g(x) = -x - 5$, find $gf(-3)$.

a. -2

b. 3

c. -5

d. -8

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